

Data Value Chain

SLIBRARY 01 · SHEET 4 / 5

Capture → store → process → analyse → act

PROFESSIONAL DEFINITION

The Data Value Chain describes the sequence of activities — capture, store, process, analyse and act — through which raw data is progressively transformed into business value, mirroring Porter's value-chain logic applied to data.

WHO DEVELOPED IT

Adapted from Michael Porter's value chain (1985) by data-management scholars and bodies such as the OECD, which formalised a data value cycle.

WHY IT WAS DEVELOPED

It was developed to help organisations see data handling as a chain of value-adding steps rather than a cost centre, and to locate where value leaks.

FIGURE 1 · DATA VALUE CHAIN



Figure 1. A schematic of the Data Value Chain as applied in BARBM312 (illustrative).

HOW IT IS USED

Each link is examined for cost, quality and value contribution; investment is directed to the link that most constrains the eventual decision.

WHEN IT IS USED

During data-strategy design, platform investment decisions, and when justifying data spend to finance.

BY WHOM IT IS USED

Chief data officers, data engineers and finance partners assessing the return on data infrastructure.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

By borrowing Porter's margin logic, it reframes data from a static asset to a flow whose value depends on the whole chain, not any single tool.

RELATED FRAMEWORKS IN THIS COURSE

DIKW Pyramid · Slibrary 1

OODA Loop · Slibrary 1

Decision-Intelligence Stack · Slibrary 1

Insight-to-Action Gap · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Porter, M.E. (1985) Competitive Advantage. New York: Free Press.

OECD (2015) Data-Driven Innovation: Big Data for Growth and Well-Being. Paris: OECD.